

Hospital User Flow

Complete Self-Explanatory Guide

Maternal Health Platform — All 8 Phases from Registration to Doctor-Patient Mapping

Overview — The 8 Phases of the Hospital Journey

Phase	Name	What happens
1	Entry and Authentication	Hospital registers once with full credentials, logs in directly every time after
2	Hospital Dashboard Overview	Single management hub — no LLM dashboard, unlike the patient side
3	Doctor Registration and Management	Hospital enters every doctor's credentials, timings, and status
4	Doctor Visibility Rules	Doctor directory is public — visible to any patient or doctor browsing
5	Patient Registration and Doctor Mapping	Hospital maps a patient to her doctor and generates her access code
6	Access Code Lifecycle	8-character code, 30-day validity, single use — links patient to hospital
7	Managing Existing Doctors and Patients	Ongoing reassignment, deactivation, and search across the hospital
8	Notifications and Edge Case Handling	Hospital-side alerts and the consolidated list of exceptions

Phase 1 — Entry and Authentication

How the hospital registers for the first time and logs in on every subsequent visit

When a hospital opens the platform for the first time it selects the Hospital profile from the same four-option landing page used by patients, doctors, and general users. Registration is a one-time event — every visit after that is a direct login.

Step 1: First-Time Registration — New Hospital

The hospital must provide the following credentials to complete registration:

- Hospital name
- Email address — used as the primary login identifier
- Password — chosen by the hospital administrator
- OTP verification — a one-time password sent to the registered email
- Address — full street/building address of the hospital
- Pincode — 6-digit postal code
- Area — locality or neighbourhood
- District — the administrative district the hospital falls under

Mandatory fields: All eight fields above are required. Registration cannot be submitted with any field left empty, and the hospital account is not created until the OTP is successfully verified.

Once the OTP is confirmed, the hospital account is created and the admin is taken directly into the Hospital Management Dashboard. This registration step happens only once for each hospital.

Step 2: Returning Hospital — Direct Login

On every visit after the first, the hospital simply enters its email and password and is taken straight to its dashboard. No OTP, no re-entry of address or pincode, no re-registration.

Edge Cases — Phase 1

Edge case	What happens
Hospital enters an email that is already registered	Registration is blocked with the message "This hospital is already registered. Please login instead."
OTP not received or expires before entry	A "Resend OTP" option appears. The new OTP is valid for 10 minutes; up to 3 resend attempts are allowed before the form must be restarted.
Pincode entered is not a valid 6-digit number	Inline validation error is shown immediately and submission is blocked until corrected.
Hospital forgets its password	"Forgot Password" sends a reset OTP to the registered email. Resetting the hospital's password has no effect on any doctor's or patient's own credentials.
A hospital chain registers multiple branches at the same address	Allowed. Each branch registers as a fully independent hospital account with its own doctors, patients, and access codes. Branches are never automatically linked.

Phase 2 — Hospital Dashboard Overview

The single management hub the hospital sees immediately after login

Unlike the patient, who sees two independent dashboards (the LLM Dashboard and the Hospital Management Dashboard), the hospital sees only one dashboard. There is no LLM chat on the hospital side — the AI assistant is a patient-only feature. Everything the hospital does lives inside a single Hospital Management Dashboard.

The dashboard is organised into the following sections, always visible:

- My Doctors — view, add, edit, or deactivate doctors registered under this hospital
- My Patients — view every patient mapped across all doctors, filterable by doctor
- Register New Doctor — add a new doctor's full credentials
- Register New Patient / Generate Access Code — map a patient to a doctor and issue her connection code
- Hospital Public Profile — the editable information every patient sees when browsing hospitals
- Notifications — hospital-side alerts described in Phase 8

Same data, one source: The Hospital Public Profile is exactly the information a patient sees when she searches for this hospital from her own dashboard (name, address, area, district, pincode, and doctor list). There is no separate copy — both sides read the same record.

Edge Cases — Phase 2

Edge case	What happens
Hospital edits its address or pincode after going live	The public profile updates instantly. Patients already connected to the hospital are unaffected.
Hospital has zero doctors registered yet	The hospital still appears in public search with its name and address. The doctor list section shows "No doctors registered yet" instead of an empty table.

Phase 3 — Doctor Registration and Management

How the hospital enters and maintains every doctor working under it

Every doctor — meaning every gynaecologist visiting or permanently attached to the hospital — must be registered by the hospital before any patient can be mapped to them.

Step 1: Registering a New Doctor

The hospital enters the following credentials for each doctor:

- Doctor's name
- Qualification
- Visiting timings at this hospital
- Number of days per week the doctor visits
- Whether the doctor is a Visiting doctor or a Permanent doctor at this hospital
- Doctor's login email
- Password — set by the hospital or auto-generated as a temporary password

Hospital clicks "Register New Doctor"

↓

Enters name, qualification, timings, days/week, permanent/visiting

↓

Enters doctor's email + sets/generates password

↓

Doctor profile created — instantly visible in the public hospital directory

↓

Doctor can now log in with these credentials into THIS hospital's portal

Portal is hospital-specific: The credentials the hospital issues let the doctor log in only into this specific hospital's portal — not into the platform generally.

Step 2: Multi-Hospital Doctors — Independent Credentials Per Hospital

The same real-world doctor may visit more than one hospital. When that happens, each hospital registers him completely independently — separate profile, separate login email and password, separate patient list. This mirrors the same isolation principle used on the patient side, where each hospital connection is entirely separate from every other.

Hospital	Doctor entry	Days/timings	Login credentials
Apollo Hospital	Dr. Ramesh — Permanent	Mon–Fri, 10am–4pm	ramesh@apollo portal — independent
Fortis Hospital	Dr. Ramesh — Visiting	Saturday only, 9am–1pm	ramesh@fortis portal — independent

Step 3: Editing and Deactivating a Doctor

Edits to a doctor's timings, days, or status apply immediately across the doctor directory, the patient profile view, and the doctor's own profile view. Deactivation follows a stricter path:

- The doctor's login is disabled immediately
- Existing patients keep full historical access to chat history, reports, and prescriptions — read-only
- No new appointments can be booked with a deactivated doctor
- Connected patients see a banner: "Dr. [Name] is no longer available at this hospital"
- The hospital must reassign affected patients to an active doctor for continued care (Phase 7)

Edge Cases — Phase 3

Edge case	What happens
Hospital registers a doctor with an email already used at this hospital	Registration is blocked with a duplicate-email error.
Hospital registers a doctor with zero patients	Fully allowed. The doctor sits in the directory with an empty patient list until the hospital maps patients to him later — there is no minimum.
Hospital tries to permanently delete a doctor who has active patients	Hard delete is blocked. Only deactivation is allowed, so that medical history and the audit trail are preserved.
Doctor's visiting days change mid-week (e.g., Saturday added)	Takes effect immediately for future appointment slots. Appointments already booked before the change are unaffected.

Phase 4 — Doctor Visibility Rules

Why the full doctor directory is public, on both the patient and the doctor side

Every doctor's details — name, qualification, timings, days per week, and permanent/visiting status — are visible in two places at once: the patient's profile and the doctor's own profile. This works symmetrically:

- Any patient — connected to that hospital or not — can open a hospital's public page and see the full list of registered doctors with all fields
- Any doctor, from his own dashboard, can search other hospitals and see their public doctor directories exactly the way a patient does

Browsing is not access: This visibility is read-only public browsing. It does not grant login access to that hospital's portal, and it does not make the browsing doctor staff there.

Who is viewing	What they see	Login required?
Patient — not connected	Full doctor list of any hospital	No
Patient — connected	Same list, with her assigned doctor highlighted	No, already logged into the platform
Doctor — browsing other hospitals	Full doctor list of that hospital	No
Doctor — own hospital dashboard	Full doctor list including himself	Yes, into his own hospital's portal

Edge Cases — Phase 4

Edge case	What happens
A doctor is Visiting at Hospital A and Permanent at Hospital B	Two fully independent entries appear in each hospital's directory. Nothing on either listing indicates they are the same person — consistent with the per-hospital isolation used everywhere else in the platform.
Hospital deactivates a doctor	He is removed from the public directory instantly. He remains visible only inside the historical records of patients who were already connected to him — not in any fresh public search.

Phase 5 — Patient Registration and Doctor Mapping

How the hospital connects a patient to the specific doctor she is consulting

The hospital is responsible for mapping every patient to the doctor she is consulting under them. Importantly, the hospital never creates the patient's login account — a patient always owns and creates her own account independently, exactly as described in the Patient User Flow. What the hospital creates is a mapping record plus a one-time access code that links her existing (or soon-to-be-created) account to this hospital and doctor.

Step-by-Step Flow

```

Hospital opens the doctor's profile → clicks "Add Patient"
    ↓
Enters patient's full name, phone number, and confirms assigned doctor
    ↓
System auto-generates an 8-character access code (e.g., BLOOM-7829)
Valid for 30 days · single use
    ↓
Hospital staff hands the code to the patient physically at her visit
    ↓
Patient enters the code on her own dashboard → "Connect to this Hospital"
    ↓
Her account is linked — chat, reports, appointments, and prescriptions unlock immediately
  
```

Account ownership stays with the patient: The hospital's registration step only creates a mapping record and an access code. It never creates or controls the patient's login credentials — those remain entirely hers.

Doctor With Zero Patients — Adding Patients Later

A hospital can register a doctor with no patients consulting him yet, exactly as required. The doctor's profile simply sits with an empty patient list. The hospital can click "Add Patient" under his profile at any time afterward — there is no limit and no minimum patient count to keep a doctor active.

Edge Cases — Phase 5

Edge case	What happens
Staff forgot to hand the code to the patient, or she lost it	The hospital dashboard still shows the code as "Unused." Staff can view, reprint, or resend it anytime within its 30-day window.
Same patient registered twice by mistake	Hospital searches by phone number before creating a new mapping. If a duplicate is found, the existing record is reused or edited instead of creating a second one.
Patient needs to move from Doctor A to Doctor B (referral, doctor left, etc.)	Handled by the Reassign action in Phase 7, which updates the doctor mapping directly. No new access code is needed since the hospital connection already exists.
Patient never uses the code within 30 days	The code auto-expires. The hospital regenerates a fresh one on request — the old code is permanently invalidated.

Phase 6 — Access Code Lifecycle

The single mechanism that links a hospital-registered patient to her own account

The access code is the same mechanism described from the patient's side in Patient User Flow Phase 3: an 8-character alphanumeric code, in the format HOSPITALPREFIX-#### (for example BLOOM-7829), valid for 30 days from generation, and single-use — it is marked used the moment the patient enters it successfully.

Code status on hospital dashboard	Meaning	Hospital action available
Active / Unused (green)	Generated, not yet entered by the patient	Reprint, resend, or wait
Used (grey)	Patient successfully connected	None needed — mapping is now visible under the doctor
Expired (red)	30 days have passed, still unused	Regenerate — creates a new code and permanently invalidates the old one

Edge Cases — Phase 6

Edge case	What happens
Patient enters a valid code while browsing the wrong hospital's connect prompt	Blocked with "This code does not belong to this hospital." The code is scoped to a specific hospital and doctor pairing and only works from that hospital's connect screen.
Hospital regenerates a code while the patient's entry is mid-submission	The old code is invalidated the instant regeneration happens. Any in-flight submission on the old code fails with "Code no longer valid — request a new one from your hospital."

Phase 7 — Managing Existing Doctors and Patients

The hospital's ongoing, day-to-day management operations

Beyond the initial registration steps, the hospital dashboard supports continuous management:

- View all doctors with a live count of patients mapped to each
- Open any doctor to see his complete patient list
- Reassign a patient from one doctor to another within the same hospital
- Deactivate a doctor, following the rules set out in Phase 3
- Search patients by name or phone number across the entire hospital
- Check any patient's access code status at a glance

Edge Cases — Phase 7

Edge case	What happens
Hospital deactivates a doctor who still has patients with pending appointments	Pending appointments are auto-flagged "Doctor unavailable — please reschedule." The hospital is prompted to reassign those patients before deactivation is finalised.
Two staff members edit the same doctor's profile at the same time	Last save wins. There is no built-in conflict merge — this is a documented limitation of the current prototype.
Hospital needs separate logins for different staff (receptionist, admin)	Out of scope for this prototype. A single hospital login is shared by all staff at that hospital.

Phase 8 — Notifications and Edge Case Handling

What the hospital is alerted about, and how

Notification type	What triggers it	Delivery method
New patient connected	A patient successfully enters her access code	Push notification on the hospital dashboard
Appointment booked	A patient books an available slot with one of the hospital's doctors	Push notification
Critical vitals alert (patient side)	A connected patient's vitals cross a danger threshold and her sharing toggle is ON for that doctor	Delivered to the doctor's own dashboard — the hospital account itself is not separately alerted, consistent with per-doctor scoping
Access code expiring soon	3 days remain before a generated code's 30-day validity ends, still unused	Push notification reminding staff to hand it over or regenerate

Design Principles Carried Over From the Patient Flow

- Per-hospital isolation — a doctor working at two hospitals is two fully independent records, never linked
- One-time OTP registration, direct login every time after

- Access codes are single-use and time-bound, exactly like the patient's hospital-connect code
- Data ownership stays with the account holder — the hospital maps patients to doctors, but never owns a patient's login credentials
- Deactivate, never hard-delete, whenever medical or audit history exists

Complete Hospital Flow Summary

Phase	Name	Key rule
1	Entry and Authentication	Register once with 8 mandatory fields + OTP. Direct login every time after.
2	Hospital Dashboard Overview	One single management dashboard — no LLM chat on the hospital side.
3	Doctor Registration and Management	Full credentials required per doctor. Deactivate, never delete, once patients exist.
4	Doctor Visibility Rules	Doctor directory is fully public — visible to any patient or doctor, no login needed to browse.
5	Patient Registration and Doctor Mapping	Hospital maps patients to doctors and issues access codes. Never owns patient login credentials.
6	Access Code Lifecycle	8-character code, 30-day validity, single use, scoped to one hospital-doctor pairing.
7	Managing Existing Doctors and Patients	Reassignment and deactivation available anytime; single shared hospital login in this prototype.
8	Notifications and Edge Case Handling	Hospital alerted on new connections, bookings, and expiring codes — not on patient vitals directly.

THE AUDIT WHAT THE HOSPITAL SEES:

Event	What the hospital sees	Source
Appointment requested	Patient name → Doctor name → requested slot → status (pending / confirmed / declined)	Triggered when patient books via Book Appointment
Video consultation requested	Patient name → Doctor name → scheduled time → status	Triggered when patient books a video slot
Medicine ordered	Patient name → prescribing doctor → order status (placed / dispatched / delivered) → amount	Triggered from Order Medicines (Razorpay flow)
Doctor attendance	Doctor name → patient → scheduled time → Attended / No-show (doctor) / No-show (patient) / Completed	Derived from video session join logs

Design Mechanics

- Each event is generated by the backend the same way patient notifications already are — the hospital audit feed is simply one more subscriber to that event alongside the patient's push and SMS pipeline. No new data source is required, just a fan-out to an additional listener.
- Stored as a `hospital_audit_log` table with fields: `event_type`, `patient_id`, `doctor_id`, `hospital_id`, `timestamp`, `status` — one row per event, filterable by doctor or by patient in the hospital's dashboard.
- Attendance tracking requires the video session (Zegocloud) to emit join and leave timestamps — either via a webhook if the plan supports it, or client-side join pings sent to the backend at call start.

How Attendance Status is Determined

At the scheduled appointment time plus a grace window (5–10 minutes):

- Both doctor and patient joined → Completed
- Doctor did not join, patient did → No-show (doctor)
- Patient did not join, doctor did → No-show (patient)
- Neither joined → Missed

Delivery — Real-Time vs Queryable Log

Routine events — appointment requests, video consultation requests, medicine orders — populate the audit log for later review and do not need to interrupt the hospital with a live notification for every occurrence.

Important: Doctor no-shows are the one event where hospital administration may want to act the same day, so this is pushed to the hospital in real time using the existing notification pipeline, while all other events remain part of the queryable audit log.

Doctor User Flow

Complete Self-Explanatory Guide

Maternal Health Platform — All 8 Phases from Personal Registration to Patient Vitals

Overview — The 8 Phases of the Doctor Journey

Phase	Name	What happens
1	Entry and Authentication	Doctor registers once with his OWN personal email — never with hospital-issued credentials
2	Doctor Dashboard Overview	A single dashboard built around a public hospital search bar
3	Browsing the Public Hospital Directory	Search any hospital, view its doctors — never its patients — no unlock needed
4	Unlocking Hospital Access	Enter the hospital-issued credentials once to unlock that hospital's patient list
5	Viewing the Patient List	See every patient mapped to him, scoped strictly to that one hospital
6	Opening a Patient Record	Name, age, gender, and registration date appear on click
7	Viewing Vitals — Live, Daily, Weekly, Monthly	Four separate views, read-only, governed by the patient's own sharing toggle
8	Multi-Hospital Access and Notifications	Independent unlocks per hospital, plus doctor-side alerts

Phase 1 — Entry and Authentication

The doctor's own personal account — completely independent of any hospital

When a doctor opens the platform for the first time he selects the Doctor profile from the same four-option landing page used by patients, hospitals, and general users. He registers using his own personal identity — never the credentials a hospital later issues him.

Step 1: First-Time Registration — Personal Account

Registration requires:

- Personal email address — his own, not a hospital-provided one
- Password — chosen by the doctor himself
- OTP verification — sent to his personal email to confirm it

Not the hospital's credentials: This registration step uses the doctor's own personal email and password. It has nothing to do with any hospital yet — it simply creates his identity on the platform, exactly the way a patient creates hers.

Once the OTP is verified, his account is created and he lands directly on his dashboard. This registration happens only once.

Step 2: Returning Doctor — Direct Login

On every visit after the first, he logs in with just his personal email and password. No OTP, no re-verification.

Edge Cases — Phase 1

Edge case	What happens
Doctor tries to register with an email already used on the platform	Blocked with "This email is already registered. Please login instead."
Doctor mistakenly enters hospital-issued credentials at this personal registration screen	The system does not recognise hospital-issued credentials here — they are only ever entered later, inside Phase 4, to unlock a specific hospital. Registration proceeds only with a genuine personal email.
OTP not received or expires	"Resend OTP" available; new OTP valid for 10 minutes, up to 3 resend attempts.
Doctor forgets his personal password	"Forgot Password" resets it via his personal email. This has no effect on any hospital-issued credentials he holds separately.

Phase 2 — Doctor Dashboard Overview

A single dashboard built around one search bar

After logging in, the doctor sees one dashboard — there is no split between an LLM area and a management area the way there is for a patient. The centrepiece is a search bar listing every hospital registered on the platform.

- Search Hospitals — the public directory, open to browse freely
- My Unlocked Hospitals — the hospitals where he has entered hospital-issued credentials and can see patients (empty until Phase 4 is completed at least once)

Edge Cases — Phase 2

Edge case	What happens
Doctor has never unlocked any hospital	He can still use the search bar and browse every public hospital directory indefinitely. "My Unlocked Hospitals" simply shows "No hospitals unlocked yet."

Phase 3 — Browsing the Public Hospital Directory

What any doctor can see about any hospital — with no unlock at all

Using the search bar, the doctor can look up any hospital registered on the platform and view its public page — the same page a patient sees when browsing hospitals, and the same directory described in the Hospital User Flow's Doctor Visibility Rules.

What is visible on any hospital's public page, without unlocking anything:

- Hospital name, address, area, district, and pincode
- The complete list of doctors registered at that hospital
- For each doctor — name, qualification, visiting timings, days per week, and permanent/visiting status

Patients are never visible here: No matter which hospital he searches — including one he is himself registered at — this public browsing view never shows any patient names or patient counts. It shows only doctor-level information.

Edge Cases — Phase 3

Edge case	What happens
Doctor searches for a hospital he is registered at, but has not unlocked yet	He sees the same public page everyone else sees, including his own entry in the doctor list. His patient list stays hidden until he completes Phase 4.
Doctor searches for a hospital with zero doctors registered	Hospital's basic details still appear; doctor list shows "No doctors registered yet," consistent with the Hospital User Flow.

Phase 4 — Unlocking Hospital Access

Entering the hospital's credentials to reveal his own patient list at that hospital

Suppose the doctor works as a visiting doctor at XYZ Hospital. To see the patients consulting him there, he must unlock that specific hospital using the credentials XYZ Hospital issued him when it registered him (Hospital User Flow, Phase 3). This unlock is entered from inside his own personal dashboard — it does not create a second account.

```

Doctor opens XYZ Hospital's public page from the search bar
↓
Clicks "Unlock My Patient Access"
↓
Enters the email + password XYZ Hospital gave him at registration
↓
Credentials verified against XYZ Hospital's records
↓
XYZ Hospital moves into "My Unlocked Hospitals" permanently
↓
His patient list at XYZ Hospital becomes visible

```

One-time, per hospital: Exactly like a patient's access code, this unlock happens once per hospital. On every future login, XYZ Hospital already shows as unlocked — he goes straight to his patient list there without re-entering anything.

Edge Cases — Phase 4

Edge case	What happens
Doctor enters the wrong hospital-issued password	"Incorrect credentials. Please check with your hospital." He may retry.
Doctor tries to unlock a hospital he was never registered at	Verification fails — there is no matching doctor record to check the credentials against.
Hospital deactivates the doctor (Hospital User Flow, Phase 3, Step 3) after he has already unlocked it	His unlock is revoked immediately. XYZ Hospital disappears from his active patient-access list. He sees a banner: "Your access at XYZ Hospital has been deactivated by the hospital." His personal account and any other unlocked hospitals are unaffected.
Doctor loses or forgets his hospital-issued password after unlocking once	Not needed again for that hospital, since the unlock is remembered permanently. If the hospital ever resets his credentials, the old unlock still stands — hospital-side credential resets do not force a re-unlock.

Phase 5 — Viewing the Patient List

Who appears under him, scoped strictly to one hospital at a time

Once unlocked, XYZ Hospital's section of his dashboard shows the list of patients mapped to him there — exactly the mapping the hospital created in its own Patient Registration and Doctor Mapping phase. If he also works at a second hospital, that hospital has its own completely separate patient list.

Edge Cases — Phase 5

Edge case	What happens
Doctor has zero patients at a hospital he just unlocked	List shows "No patients assigned yet" — matches the hospital-side case of a freshly registered doctor.
Hospital maps a new patient to him after he already unlocked the hospital	She appears in his patient list automatically the next time he opens it — no need to unlock again.
Hospital reassigns one of his patients to a different doctor	She disappears from his list immediately. Historical chat, reports, and prescriptions he already exchanged with her remain in his own records for reference, but no further live data is shared.

Phase 6 — Opening a Patient Record

What appears the moment he clicks on a patient's name

Clicking a patient in his list opens her record. In addition to her name, the following become visible:

- Age
- Gender
- Registration date — when she was first mapped to him under this hospital

From here a "View Vitals" button leads into her vitals data, covered in Phase 7.

Edge Cases — Phase 6

Edge case	What happens
Patient shared her pre-connection self-recorded history with him (Patient User Flow, Phase 6)	Her record and vitals include everything from day one, marked with source "self" for the pre-connection entries.
Patient chose to share only from her connection date onward	Her record shows registration date correctly, but no vitals or history exist before that date — nothing to display for that period.
Patient's age or gender field is left blank at registration	Field displays "Not provided" rather than a blank or broken cell.

Phase 7 — Viewing Vitals: Live, Daily, Weekly, Monthly

Four separate, read-only views inside the patient's record

Clicking "View Vitals" opens four distinct sections, matching the same structure the patient sees on her own dashboard (Patient User Flow, Phase 4):

View	What it shows	Updated how often
Live readings	Current value of each parameter, streamed over WebSocket	Per-parameter interval — HR/SpO ₂ every 30s, BP hourly, etc.

View	What it shows	Updated how often
Daily vitals	Today's readings and a same-day summary	Throughout the day
Weekly averages	Average per parameter over the last 7 days	Updated daily at midnight
Monthly averages	Average per parameter over the last 30 days	Updated daily at midnight

Read-only, and never his to control: The doctor can only view vitals — he cannot edit, delete, or trigger a reading. Whether his view is live at all is entirely controlled by the patient's own per-doctor sharing toggle (Patient User Flow, Phase 5).

Edge Cases — Phase 7

Edge case	What happens
Patient has her live-sharing toggle OFF for this doctor	Live readings section shows "Patient has paused live vitals sharing." Daily, weekly, and monthly historical data recorded before the toggle was switched off remain fully visible — the toggle affects only the live feed.
Patient has never connected her wearable	All four sections show "No vitals recorded yet" instead of empty charts.
Patient disconnects her wearable device entirely	Live readings show "Device offline," with the last known values greyed out. Daily/weekly/monthly aggregates already recorded remain visible.
Wearable is placed on a table (off-body state)	Live readings go flat or show a gap with a "Sensor Off-Body / No Signal" warning — never mistaken for a health crisis.
A genuine crisis spike occurs on a patient with the toggle ON	A red alarm banner appears instantly on his dashboard, delivered the same moment the patient's own critical-vitals alert fires.

Phase 8 — Multi-Hospital Access and Notifications

How everything scales when a doctor works at more than one hospital

A doctor's personal account is singular, but his access to patients is not. If he is a visiting doctor at XYZ Hospital and a permanent doctor at ABC Hospital, he unlocks each one independently (Phase 4), using the separate credentials each hospital issued him. His dashboard then shows two fully separate sections — XYZ Hospital's patient list and ABC Hospital's patient list — with no crossover between them.

Hospital	His role there	Unlock status	Patients visible
XYZ Hospital	Visiting	Unlocked with XYZ's credentials	Only XYZ Hospital's mapped patients
ABC Hospital	Permanent	Unlocked with ABC's credentials	Only ABC Hospital's mapped patients

Doctor-Side Notifications

Notification type	What triggers it	Delivery method
Critical vitals alert	A patient with toggle ON crosses a danger threshold	Push notification + red banner, simultaneous with the patient's own alert
New patient assigned	A hospital maps a new patient to him at an already-unlocked hospital	Push notification

Notification type	What triggers it	Delivery method
Access deactivated	A hospital deactivates his profile there	Push notification and banner inside that hospital's section
Tele-consultation message	A patient sends him a message in the shared chat	Push notification

Edge Cases — Phase 8

Edge case	What happens
Doctor is the same real person at two hospitals, but each hospital registered him with slightly different spelling/qualification text	No reconciliation happens automatically. Each hospital's entry is authoritative for its own portal, matching the per-hospital isolation used throughout the platform.
Doctor unlocks a hospital, then that hospital's own login page or credentials change on the hospital's side later	The doctor's existing unlock is unaffected unless the hospital explicitly deactivates him. Hospital-side password resets do not require the doctor to re-enter anything.
Doctor tries to unlock the same hospital twice	Not necessary — once unlocked, the option changes to "View Patients" instead of "Unlock," and re-entering credentials is not required again.

Design Principles Carried Over From the Patient and Hospital Flows

- One-time personal registration with OTP, direct login every time after — same pattern as the patient
- A separate, hospital-issued credential acts purely as a one-time unlock key, never as a second account — parallel to the patient's single-use access code
- Per-hospital isolation — patient lists, and now doctor access, never cross between hospitals
- Vitals views are read-only and fully governed by the patient's own sharing toggle, never by doctor action
- Deactivation revokes access instantly but preserves historical records, consistent with the hospital's deactivate-not-delete rule

Complete Doctor Flow Summary

Phase	Name	Key rule
1	Entry and Authentication	Personal email + OTP, once. Direct login every time after. Never hospital-issued credentials here.
2	Doctor Dashboard Overview	One dashboard, centred on a public hospital search bar.
3	Browsing the Public Hospital Directory	Any hospital's doctor list is public. Patient data is never shown here.
4	Unlocking Hospital Access	Hospital-issued credentials unlock that hospital once, permanently — not a second login.
5	Viewing the Patient List	Strictly scoped per hospital. Zero patients is a valid, supported state.
6	Opening a Patient Record	Name, age, gender, and registration date shown on click.
7	Viewing Vitals	Live, Daily, Weekly, Monthly — read-only, governed entirely by the patient's toggle.

Phase	Name	Key rule
8	Multi-Hospital Access and Notifications	Independent unlock per hospital; critical alerts fire the instant a toggle-ON patient crosses threshold.

Patient User Flow

Complete Self-Explanatory Guide

Maternal Health Platform — All 7 Phases from Entry to Notifications

Overview — The 7 Phases of the Patient Journey

Phase	Name	What happens
1	Entry and Authentication	Patient opens app, chooses profile, registers or logs in — one time only
2	Two Main Dashboards	After login — LLM Dashboard and Hospital Management Dashboard split
3	Hospital Search and Connection	Browse any hospital publicly, enter access code once to connect
4	Wearable Device Connection	Independent button spawns simulation thread — live vitals begin
5	Vitals Sharing Control	Toggle per doctor controls whether live vitals reach their dashboard
6	Self-Recording Before Hospital	Collect own vitals before connecting — choose to share when ready
7	Notifications	Push and SMS notifications for all critical and routine events

Phase 1 — Entry and Authentication

How the patient enters the platform and logs in for the first and subsequent times

When a patient first opens the application she sees the landing page with four profile options. She selects the one that matches her role. A maternal patient selects the maternal patient profile. A general user selects the general user profile. The doctor and hospital each have their own separate profiles.

Step 1: Landing Page — Profile Selection

The patient sees four clearly labelled profile options on the landing page:

- Maternal Patient — for pregnant women using the platform for pregnancy monitoring and guidance
- Doctor — for gynecologists and other doctors using the platform to monitor their patients
- Hospital — for hospital organisations to register and manage their doctors and patients
- Other / General User — for any user who wants to use the platform without a specific medical context

Note: A maternal patient and a general user follow the same authentication and dashboard flow. The only difference is the LLM behaviour. Both have access to hospital browsing, wearable monitoring, and vitals tracking.

Step 2: First Time Registration — New User

If the patient is using the platform for the first time she registers herself. No hospital involvement is needed for this step. She creates her own account independently.

Registration fields:

- Email address — used as the primary login identifier
- Password — chosen by the patient
- OTP verification — a one-time password is sent to her email to confirm it is valid

After OTP verification her account is created and she is taken directly into her dashboard. This registration step happens ONLY ONCE. Every subsequent visit she logs in directly without OTP.

Step 3: Returning User Login — Direct Access

On every visit after the first, the patient simply enters her email and password and is taken straight to her dashboard. No OTP. No additional verification. No re-registration. Direct access every time.

Important: The platform remembers her session. On most devices she will remain logged in between visits. She only needs to enter credentials again if she explicitly logs out or changes her device.

Phase 2 — Two Main Dashboards

The two independent areas the patient sees immediately after logging in

After login the patient sees two clearly separated dashboards. These are two completely independent areas of the platform. She can use one without the other. They do not interfere with each other.

Dashboard 1: LLM Dashboard — Lifestyle Companion

The LLM Dashboard is a private AI chat interface. It is available to all users — maternal patients and general users alike. It is the first option visible after login.

What a maternal patient can do here:

- Diet and nutrition advice based on her trimester and health conditions
- Exercise and physical activity recommendations
- Emotional and mental wellbeing support and active listening
- Sleep and rest guidance
- Hydration and daily water intake targets
- Trimester-specific lifestyle tips proactively shared
- Baby kick tracking reminders — minimum 10 kicks per day
- General pregnancy questions answered using a decision framework
- Report explanation — LLM translates medical terms without advising treatment
- Ask the Doctor question generator — generates questions for patient to ask her doctor
- On-demand vitals analysis — describes what vitals show, never advises how to fix them
- Companion conversations — warm friendly chat for emotional support and lighthearted moments

What a general user can do here:

- Chat about general topics — stocks, sales, personal interests, everyday questions
- The LLM responds freely to non-health topics
- For any health-related question — LLM shows notice: "This assistant is designed for maternal health guidance only"

- General users cannot bypass maternal health restrictions regardless of how the question is framed

Critical rule: The LLM communicates ONLY with the patient. It has NO connection to the doctor whatsoever. It does not send notifications, alerts, or messages to the doctor at any point. The doctor never sees LLM conversations. The responsibility of reaching the doctor always lies with the patient.

Dashboard 2: Hospital Management Dashboard — Clinical Hub

The Hospital Management Dashboard is where the patient manages everything related to her hospital connections, wearable device, and live vitals. It contains three independent sections that are always visible:

- My Wearable Device — connect or disconnect the simulation, view live vitals
- My Connected Hospitals — access connected hospitals, doctors, reports, appointments, chat, prescriptions
- All Registered Hospitals — browse and view any hospital on the platform publicly

Key point: Connecting to a hospital and connecting the wearable device are two completely separate actions. One does not trigger the other. Both are independent.

Phase 3 — Hospital Search and Connection

How the patient finds a hospital, browses doctors, and connects using her access code

The patient does not need to connect to a hospital to use the platform. She can browse hospitals, use the LLM, and monitor her own vitals entirely independently. Hospital connection is optional and can be done at any time.

Step 1: Browsing Any Hospital — No Code Needed

From the Hospital Management Dashboard the patient types any hospital name in the search bar. The hospital appears in results. She clicks on it and sees the hospital's public page.

What is publicly visible to any patient — no connection required:

- Hospital name, full address, area, district, and pincode
- Complete list of all doctors registered in that hospital
- Each doctor's name, qualification, visiting timings, days per week, and whether permanent or visiting

Important: A patient can view the full doctor list of any hospital — even one she is not consulting at. This is public information on the platform. No code. No login to hospital. No verification needed.

Step 2: Connecting to a Hospital — The Access Code

When the patient decides to connect to a specific hospital she clicks "Connect to this Hospital". A single input field appears asking for her hospital access code.

What is the access code:

- An 8-character alphanumeric code auto-generated by the hospital system when they register her as a patient
- Format example: BLOOM-7829
- Valid for 30 days from generation
- Single use — marked as used the moment she enters it successfully
- Given to her physically by hospital staff at her first in-person consultation visit

The three possible outcomes when she enters the code:

Outcome	Condition	What happens
Valid ✓	Code correct + within 30 days + unused	Connection established immediately. Code marked as used. She never needs it again.
Wrong code ✗	Incorrect characters entered	Message shown: "Incorrect code. Please check and try again." She can retry.
Expired ⌚	30 days have passed since generation	Message shown: "Code expired. Contact your hospital for a new code." Hospital regenerates.

Step 3: After Successful Connection — What She Gets

One code entry unlocks everything for that hospital simultaneously. She never needs to enter credentials again for that hospital. All of the following become available immediately:

- Her assigned doctor's full details visible in her dashboard
- Tele-consultation chat — direct human-to-human messaging with the doctor
- My Reports — medical reports uploaded by the hospital, viewable as PDF
- Book Appointment — see available slots from the doctor and book directly
- Video Consultation — available at appointment time via Zegocloud
- My Prescriptions — prescriptions written by the doctor with medicine details
- Order Medicines — auto-fills cart from prescription and processes payment via Razorpay UPI
- Live vitals sharing toggle for this doctor — controls whether doctor sees her live vitals

Next visit: On every subsequent login Apollo Hospital shows as Connected automatically in her dashboard. No code entry ever again. She goes directly to her doctor, reports, chat, and appointments.

Step 4: Connecting to Multiple Hospitals Simultaneously

A patient can connect to more than one hospital at the same time. For example she may consult Apollo Hospital for regular checkups and Fortis Hospital for a specialist. Each hospital connection is completely separate.

- Each hospital has its own doctor card, chat room, reports section, appointment page, and prescriptions
- Vitals are broadcast to all connected doctors whose toggle is ON simultaneously
- WebSocket rooms are isolated per patient-doctor pair — data never crosses between hospitals
- Patient controls each hospital connection and each vitals toggle independently

Phase 4 — Wearable Device Connection

How the patient starts and stops her vitals simulation — independent of hospital connection

The wearable connection is completely independent from hospital connection. The patient does not need to be connected to any hospital to connect her wearable. She can monitor her own vitals entirely without a doctor.

In this prototype the "wearable device" is a Python simulation script running on a cloud server. It acts as the software equivalent of a real IoT wearable device. When the patient clicks "Connect Wearable Device" she is triggering a backend API call that spawns a simulation thread — not pairing a physical Bluetooth device.

For evaluators: In a real deployment this button would initiate Bluetooth pairing with the physical wearable device which then connects to Wi-Fi and begins transmitting. In this prototype the button triggers an on-demand Python simulation thread on the cloud server that mimics the exact same data flow — MQTT publish, backend processing, WebSocket delivery — that a real device would produce. The architecture is identical. Only the data source differs.

Connecting the Wearable:

The patient goes to the My Wearable Device section in her Hospital Management Dashboard.

```
Status: ○ Not connected
Charts: empty

Patient clicks "Connect Wearable Device"
↓
Loading spinner: "Starting your health monitor..."
↓
Backend spawns simulation thread for Patient P123
Thread begins generating:
HR · SpO2 · BP · Glucose (CGM) · Sleep · Steps · Kicks · CO2
↓
Status: ● Connected — Live vitals active
Green indicator appears on dashboard permanently
Vitals panel populates with live values
```

What Vitals She Sees:

Once connected the patient sees her vitals in three views simultaneously:

View	What it shows	Updated how often
Live readings	Current value of each parameter	Per parameter interval — see table below
Daily vitals	Today's readings and daily summary	Updated throughout the day

View	What it shows	Updated how often
Weekly averages	Average per parameter over last 7 days	Updated daily at midnight
Monthly averages	Average per parameter over last 30 days	Updated daily at midnight

Vital Parameters and Their Update Intervals:

Parameter	Real interval	Demo interval	Edge method
Heart Rate	Every 30s with deadband	Every 30s	Deadband ± 10 bpm — suppress if no change
SpO ₂ (Oxygen)	Every 30s with deadband	Every 30s	Deadband $\pm 2\%$ — suppress if no change
Blood Glucose (CGM)	Every 1–5 mins with deadband	Every 60s	Deadband ± 5 mg/dL — suppress if no change
Blood Pressure	Every 1 hour	Every 30s in demo	Moving window average of 3 readings
Sleep Quality	Once daily at midnight	Every 90s in demo	Local aggregation — daily summary packet
Physical Activity/Steps	Once daily at midnight	Every 90s in demo	Local aggregation — daily summary packet
Baby Kicks	Once or twice daily	Every 45s in demo	Patient logs — minimum 10 per day

Disconnecting the Wearable:

The patient can disconnect at any time by clicking "Disconnect Device".

```

Patient clicks "Disconnect Device"
    ↓
Backend kills simulation thread for P123
Data generation stops immediately
    ↓
Status: ○ Disconnected
Vitals panel stops updating
Last known values shown greyed out
Doctor dashboard shows: "Patient device offline"

```

Important: Logging out of the app does NOT disconnect the wearable. The simulation thread keeps running even when the browser is closed. Only clicking "Disconnect Device" stops the thread. This mirrors real-world behaviour where a wearable device keeps transmitting even when the patient's phone screen is off.

Phase 5 — Vitals Sharing Control

How the patient controls which doctor sees her live vitals — independently per doctor

The wearable connection and the vitals sharing toggle are two completely separate controls. Connecting the wearable does NOT automatically share vitals with any doctor. The patient must explicitly enable sharing per doctor using the toggle.

The Three Independent Controls:

Control	What it does	When it is set
Connect Wearable button	Starts simulation thread — patient sees her own vitals	Any time — independent
Live vitals toggle per doctor	Controls whether that doctor sees the live WebSocket feed	Any time — per doctor
Disconnect from hospital	Removes all access for that doctor permanently	Patient's choice

Toggle ON — Doctor Sees Live Vitals:

When the patient enables the toggle for a specific doctor:

```
Patient sets: Live vitals sharing — Dr. Ramesh: ● ON
↓
Backend adds Dr. Ramesh socket to WebSocket room_P123_DOCRAMESH
Vitals start flowing to his dashboard immediately
His dashboard shows green connected indicator
Doctor sees: HR · SpO2 · BP · Glucose · Sleep · Steps · Kicks
```

Toggle OFF — Doctor Sees Nothing Live:

When the patient disables the toggle for a specific doctor:

```
Patient sets: Live vitals sharing — Dr. Ramesh: ○ OFF
↓
Backend removes Dr. Ramesh socket from room_P123_DOCRAMESH
Live vitals stop flowing to his dashboard immediately
His dashboard shows: "Patient has paused live vitals sharing"
Patient still sees her own vitals normally — unaffected
Historical records still visible to doctor — only live feed paused
```

Three rules that always apply:

- Rule 1 — Toggle affects live vitals only. Historical records are always visible to the doctor regardless of toggle state.
- Rule 2 — Doctor is notified when toggled off. His dashboard shows an informational status — not an alarm.
- Rule 3 — Patient can toggle at any time. No hospital permission needed. No doctor approval needed. It is her data.

Multiple Doctors — Independent Toggle Per Doctor:

If the patient is connected to multiple hospitals with multiple doctors, each doctor has their own independent toggle:

```
Apollo Hospital — Dr. Ramesh: ● ON → receives live vitals
Fortis Hospital — Dr. Priya: ○ OFF → receives nothing live

WebSocket rooms active:
room_P123_DOCRAMESH ← active, vitals flowing
room_P123_DOCPRIYA ← inactive, no data flowing

Patient sees her own vitals on both — unaffected by either toggle
```

Phase 6 — Self-Recording Before Hospital Connection

How the patient collects her own vitals before connecting to any doctor

A patient can use the wearable and collect vitals records completely independently — without connecting to any hospital or doctor. This is a fully supported use case. Many women begin monitoring their pregnancy before their first hospital visit.

How Self-Recorded Vitals Are Stored:

When the patient collects vitals without a hospital connection, the database stores these records with the following structure:

```
vitals table row:
patient_id: P123
value: 82 bpm
recorded_at: 2026-07-25 10:30:00
doctor_id: null ← no doctor assigned yet
hospital_id: null ← no hospital connected yet
source: "self" ← patient is self-recording
```

These records are stored safely under her patient profile. They are visible to her at all times in her vitals dashboard. They are completely private — no doctor or hospital can see them yet.

When She Later Connects to a Hospital and Doctor:

When the patient later connects to a hospital and is assigned a doctor — for example after her first trimester — a one-time prompt appears asking her what she wants to share with that doctor:

```
"Would you like to share your pre-connection  
vitals records with Dr. Ramesh?  
  
[Yes — Share all past records from day one]  
[No — Share only from today onward]"
```

Patient chooses	What doctor sees
Yes — Share all past records	Doctor sees ALL vitals from day one including self-recorded history. Source field shows "self" for pre-connection records.
No — Share only from today	Doctor sees only vitals from the connection date onward. Self-recorded history stays completely private to patient.

Patient data ownership: This decision is made by the patient — not the hospital and not the doctor. Pre-connection vitals are personal records collected independently. She has full control over whether to share them.

General User — Same Self-Recording Capability:

A general (non-maternal) user can also connect the wearable and collect vitals records in exactly the same way. They see the same live vitals dashboard, weekly averages, and monthly averages. The only differences are that the LLM gives open responses instead of maternal-specific guidance, and the vitals normal ranges may differ from maternal baselines.

Phase 7 — Notifications

How the patient is kept informed about critical events and routine reminders

The patient grants push notification permission on her first login. Once granted, she receives two types of notifications — browser push notifications for all events, and SMS for critical clinical events only.

	Notification type	What triggers it	Delivery method
1	Critical vitals alert	Her own vitals cross a danger threshold — BP above 140, SpO ₂ below 94, HR above 120, glucose below 70 or above 140	Push notification + SMS both sent simultaneously
2	Doctor message	Dr. Ramesh sends a message in the tele-consultation chat	Push notification
3	Appointment reminder	An appointment with Dr. Ramesh is approaching	Push notification 1 hour before — SMS also sent
4	Daily kick count reminder	She has not logged her kick count for the day	Push notification once daily in the afternoon
5	Medicine order status	Order placed, dispatched, or delivered	Push notification at each status change
6	New report uploaded	Apollo Hospital uploads a blood test, scan, or other report	Push notification with report type and date

How Push Notifications Work

Browser push notifications via the Web Push API work even when the browser is closed or minimised. The patient grants permission once on first login. After that, notifications appear on her device even if the application tab is not open.

- On first login — browser asks: "Allow notifications from this app?" — patient clicks Allow
- After that — notifications arrive automatically without the app being open
- Clicking any notification opens the app directly to the relevant section

Example: Patient receives notification: "New report available — Apollo Hospital — Blood Test — 25 July 2026 — Tap to view." She taps it. App opens directly to the My Reports section under Apollo Hospital.

Critical Vitals Alert — Special Handling

Critical vitals alerts are the most important notification type. They are delivered both as push notification and SMS simultaneously to ensure the patient receives them even if her phone is on silent or the app is not installed on that device.

```

Patient's BP reading: 155/98 mmHg
↓
Backend detects threshold breach (systolic > 140)
↓
Push notification sent:
"□ Your blood pressure reading is elevated at 155/98 mmHg.
  This is above the safe threshold of 140/90.
  Please contact your doctor immediately."
↓
SMS sent simultaneously to registered phone number:
"ALERT: Your BP reading is 155/98 — above safe range.
  Contact your doctor immediately."
↓
If she has a connected doctor with toggle ON:
Doctor receives alert on his dashboard simultaneously

```

Important: The LLM does NOT trigger or manage these notifications. Vitals alerts are handled entirely by the backend threshold detection system. The LLM has no connection to the notification pipeline.

Complete Patient Flow Summary

Phase	Name	Key rule
1	Entry and Authentication	Register once with email + OTP. Direct login every time after. One time only.
2	Two Main Dashboards	LLM Dashboard and Hospital Management Dashboard are completely independent.
3	Hospital Search and Connection	Browse any hospital publicly — no code. Enter BLOOM-7829 once to connect — unlocks everything permanently.
4	Wearable Connection	Completely independent from hospital connection. Spawns simulation thread. Patient always sees her own vitals.
5	Vitals Sharing Control	Toggle per doctor. Independent per doctor. Does not affect patient's own vitals view. Historical records always visible.
6	Self-Recording	Vitals stored with source = "self". Private until patient chooses to share on first hospital connection.
7	Notifications	Push for all events. SMS for critical vitals and appointment reminders. Doctor receives critical alerts simultaneously.

The Three Key Independent Controls — Summary

Control	What it is independent from	Where it lives in dashboard
Hospital access code (BLOOM-7829)	Does not auto-connect wearable. Does not auto-share vitals.	Hospital search page — enter once
Connect Wearable Device button	Does not require hospital connection. Does not auto-share vitals with doctor.	My Wearable Device section
Live vitals toggle per doctor	Does not affect patient's own vitals. Does not disconnect wearable. Historical records unaffected.	Inside connected hospital card